Skellefteå 2021-08-12

Project: 84691 Delhi Fire Station

To whom it may concern,

Door type	:VL3116
Dimensions	:CWxCH 9000x5400

We certify that the above calculated door have been designed to withstand a differential pressure of 1.15kPa x beam safety factor of 1,5.

The doors should be installed and serviced by authorized installers certified by ASSA ABLOY

Best regards



Oskar Vestermark

Technical Manager

ASSA ABLOY ENTRANCE SYSTEMS

MEGADOOR

ASSA ABLOY

Calculation of uniformly loaded intermediate beam

Project: **Delhi Fire Station**
Project number: 84691
Date: **2021-08-12**
Name: Mats Brännström

Nominal differential pressure: $P := 1150 \cdot \frac{N}{m^2}$

Fabric contribution: $cf := 1.056$

Reduction factor due to profile rotation: $rf := 0.970$

Actual pressure to be supported by intermediate beam: $p := \frac{P}{cf}$ $p = 1089.015 \ \frac{N}{m^2}$

Moment of inertia $I := rf \cdot 5280000 \cdot mm^4$ $I = 5121600 \ mm^4$

Support distance $l := 9000 \cdot mm$

Plastic section modulus $Z := rf \cdot 78100 \cdot mm^3$ $Z = 75757 \ mm^3$

Yield point $\sigma_y := 240 \cdot \frac{N}{mm^2}$

Modulus of elasticity $E := 70000 \cdot \frac{N}{mm^2}$

Safety factor acc. to EN 12604 $n_s := 1.5$

Allowable stress $\sigma_a := \frac{\sigma_y}{n_s}$ $\sigma_a = 160 \ \frac{N}{mm^2}$

Max allowed beam spacing $d. := \frac{\sigma_a \cdot 8 \cdot Z}{l^2 \cdot p}$ $d. = 1099.3 \ mm$

Actual spacing $d := 961 \cdot mm$

Beam stress at actual spacing $\sigma := \frac{d}{\left(\frac{8 \cdot Z}{l^2 \cdot p}\right)}$ $\sigma = 140 \ \frac{N}{mm^2}$

Skenbar förkortning
Apparent shortening
Contraction apparente

$$\Delta := \int_{0 \cdot mm}^{l} \sqrt{1 + \left(\frac{d}{dx}\left(\frac{5 \cdot d \cdot l \cdot p \cdot l^3}{384 \cdot E \cdot I} \cdot \left(1 - \frac{4 \cdot x^2}{l^2}\right) \cdot \left(1 - \frac{4 \cdot x^2}{5 \cdot l^2}\right)\right)\right)^2} \, dx - l$$

$\Delta = 17.159 \ mm$

Deflection $\delta := \frac{5 \cdot d \cdot p \cdot l^4}{384 \cdot E \cdot I}$ $\delta = 249 \ mm$